

2010376103 CSE4261 Assignment-10

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1 Generating five images passing five noise vectors/matrices drawn from a normal distribution with mean 5 and variance through the Decoder of your normal Autoencoder.

[Github code link](#)

A normal autoencoder compresses an image into a low-dimensional vector and tries to reconstruct the same image. It does not learn a structured or continuous latent space. An autoencoder has been trained in PyTorch to generate images. Figure 1b shows the output of the generated image Using normal autoencoder. Figure 1a shows the noise vectors.

- Trained on clean input images like MNIST digits.
- Encoder maps input images to a fixed-size latent vector.
- Decoder reconstructs the original image from this latent vector.

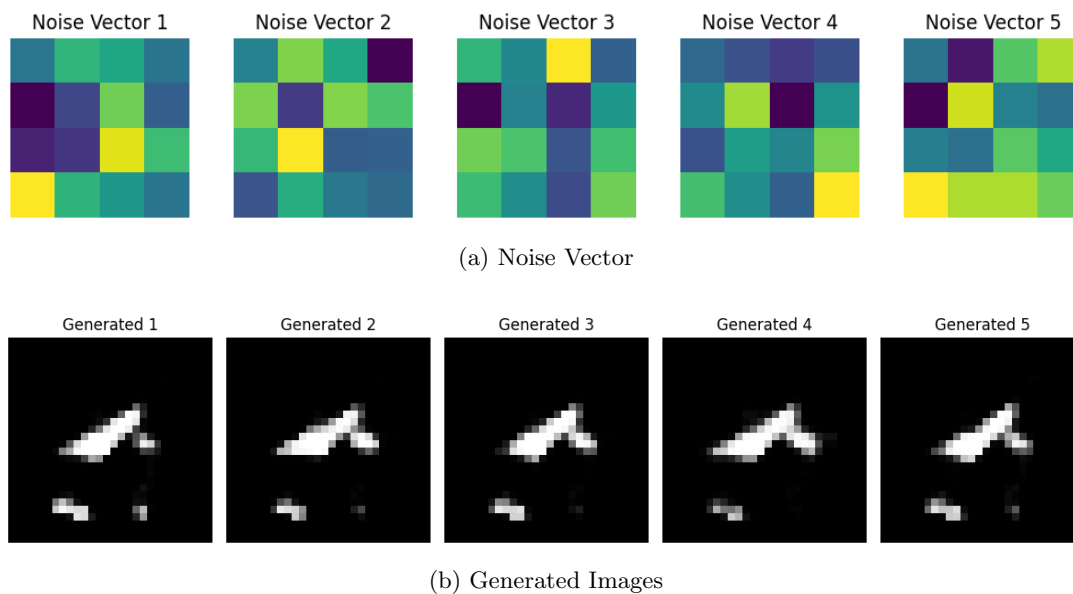


Figure 1: Generated images through the Decoder normal Autoencoder

Output when noise is passed: If you pass a random noise vector (e.g., from $\mathcal{N}(5,1)$) to the decoder of a normal AE, the output is usually meaningless. The decoder is not trained to handle such random vectors, so it produces blurry or noisy outputs. We see faint blobs, some vague digit-like shapes, or pure artifacts. These images do not resemble valid MNIST digits and vary unpredictably.

2 Generating five images passing five noise vectors/matrices drawn from a normal distribution with mean 5 and variance through the Decoder of your Denoising Autoencoder.

[Github code link](#)

A denoising autoencoder is trained to clean noisy images and recover the original clean image. We trained a denoising autoencoder to generate images while noise vector is passed through decoder of autoencoder. Figure 2b shows the output of the generated image Using Denoising autoencoder. Figure 2a shows the noise vectors.

- Trained by adding noise (mean 5) to input images.
- Learns to map noisy inputs to their clean versions.
- Encoder compresses the noisy input into a latent vector.
- Decoder reconstructs the clean image from that vector.

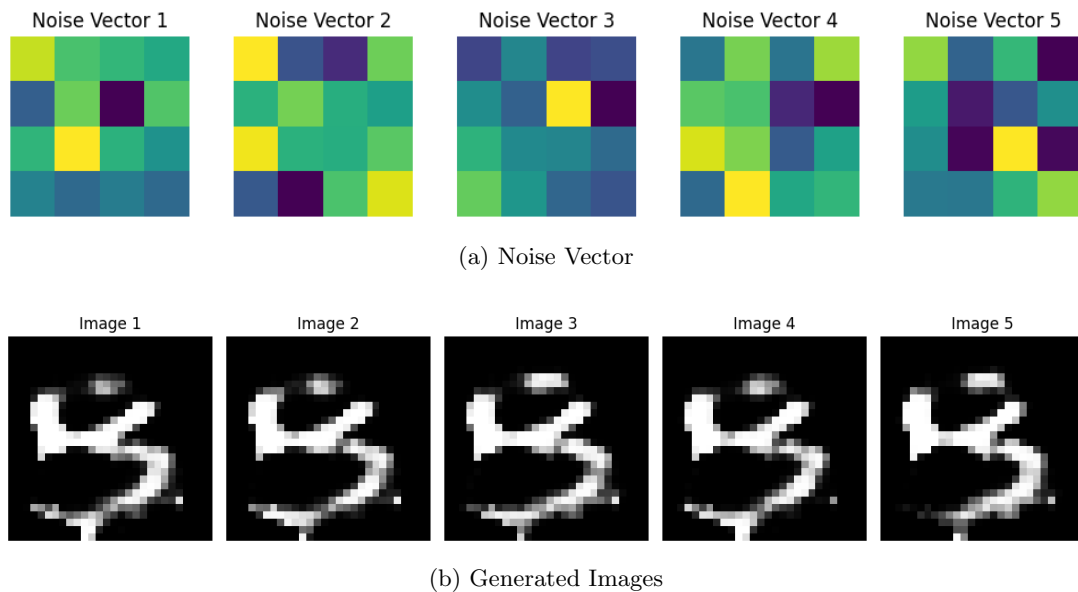


Figure 2: Generated images through the Decoder of Denoising Autoencoder

Output when noise is passed: Passing a random noise vector directly to the decoder of a DAE results in low-quality or meaningless images. The decoder expects a latent vector that comes from a noisy image passed through its encoder. Feeding random noise instead leads to output that looks like scattered pixels, ghost-like smudges, or unrecognizable shapes. But output is better than normal autoencoder. But, it fails to produce valid digits or clean structures.

3 Training and evaluating a Variational Autoencoder.

[Github code link](#)

We trained a Variational Autoencoder (VAE) on the MNIST dataset to learn a compressed, structured latent space. The model can reconstruct digits from this space and also generate new digit images by sampling from a normal distribution. A comparison between 5 original and 5 generated images shows that the VAE successfully captures the digit structure, though with slight blurriness in Figure 3.

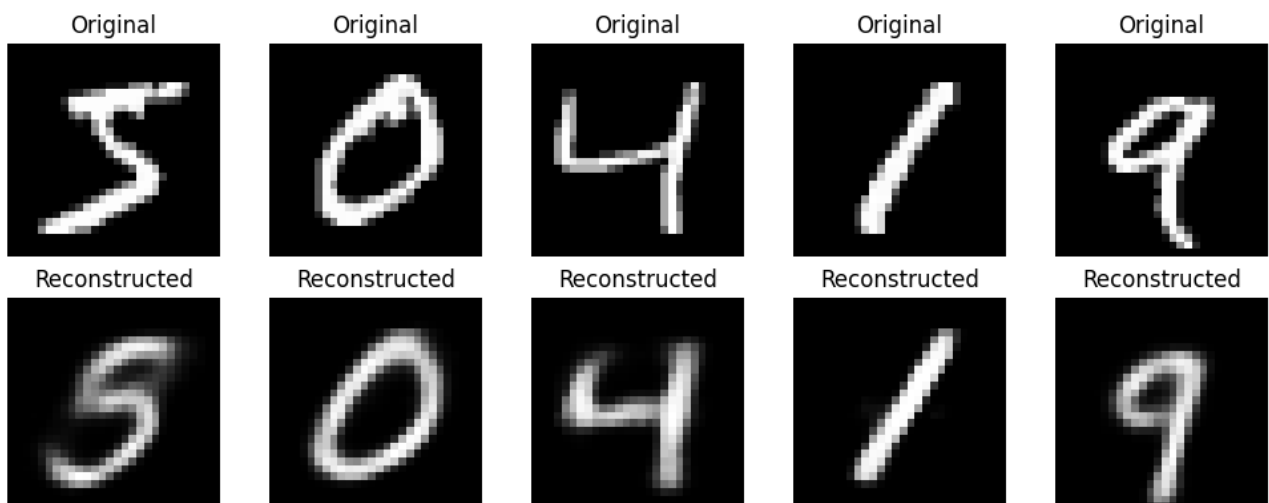


Figure 3: Reconstructed Images by VAE

4 Generating five images passing five noise vectors/matrices drawn from a normal distribution with mean 5 and variance through the Decoder of your Variational Autoencoder.

[Github code link](#)

A VAE learns a structured latent space by encoding inputs into a distribution (mean and variance). We trained a VAE to generate Images in PyTorch. Figure 4 shows the output of the generated image Using VAE, while noise vector is passed through decoder having mean 5. Figure 4 shows the output of the generated image Using VAE, while noise vector is passed through decoder having mean 0 and standard deviation 1.

- Encoder outputs mean and variance vectors for each image.
- Latent vectors are sampled from $\mathcal{N}(\mu, \sigma^2)$ during training.
- Decoder reconstructs images from these sampled vectors.
- A KL divergence loss ensures latent space is close to $\mathcal{N}(0, I)$.

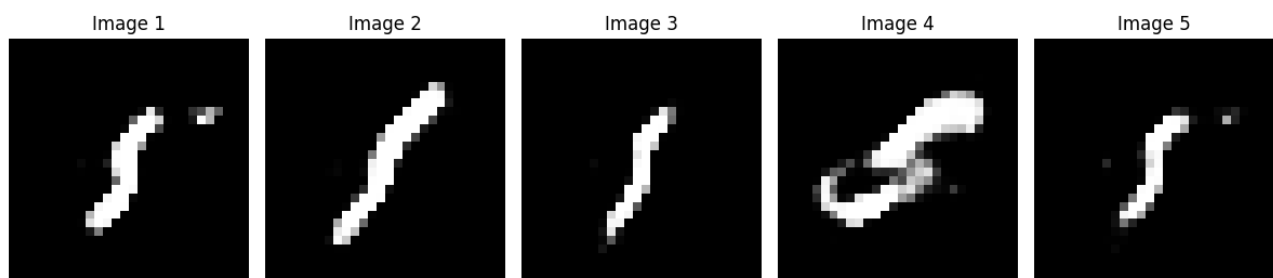


Figure 4: Noise Vector and Generated images by VAE

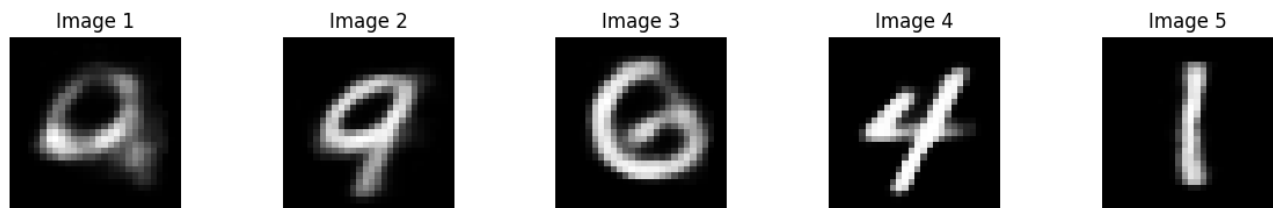


Figure 5: Generated Images from Latent Space (mean=0)

Output when noise is passed: When noise vector having mean 5 is passed through decoder of VAE, this time we can read the digit. it is clear than before in auto encoder and denoising autoencoder. but still it is blur in some places. Passing a noise vector from $\mathcal{N}(0, 1)$ to the decoder of a VAE works well. It produces clean and realistic MNIST digits. The VAE is trained to expect such noise and can map it to valid data samples. If we pass noise from $\mathcal{N}(5, 1)$ instead, the images may still look like digits but may be distorted, blurred, or look unrealistic. The further the noise is from $\mathcal{N}(0, 1)$, the less meaningful the output becomes.